Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*	(i)		$CO_2 + C \rightarrow 2CO$ award (1) for reactants and products award (1) for balancing only if reactants and products correct	2			2		
		(ii)		award (1) for any of following • limestone forms lime / quicklime • calcium carbonate forms calcium oxide / • $CaCO_3$ forms CaO • $CaCO_3 \rightarrow CaO + CO_2$ award (1) for any of following • lime / quicklime reacts with sand (to form slag) • calcium oxide reacts with silicon dioxide to form slag • $CaO + SiO_2 \rightarrow CaSiO_3$ award (1) for identification of one of the reaction types e.g. • thermal decomposition / breaks down with heat • neutralisation	3			3		
	(b)	(i)		$Fe_2O_3(s) + \mathbf{6}HCl(aq) \rightarrow \mathbf{2}FeCl_3(aq) + \mathbf{3}H_2O(l)$		1		1	1	
		(ii)		$3OH^-(aq) + Fe^{3+}(aq) \rightarrow Fe(OH)_3(s)$ award (1) for product award (1) for balancing **only** if all formulae are correct		2		2		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
(c)	(i)		high purity oxygen is used ☐ impurities are oxidised forming heat ☐ oxygen is blasted in at supersonic speed ☐ scrap steel is used in the process ✓			1	1		
	(ii)		ductility increases, hardness increases ☐ tensile strength increases, ductility increases ☐ ductility decreases, tensile strength increases ✓ hardness increases, tensile strength decreases ☐			1	1		
	(iii)		0.2 ☐ 0.6 ☐ 1.0 ✓ 1.5 ☐			1	1		
	(iv)		low carbon steel			1	1		
			Question 6 total	**5**	**3**	**4**	**12**	**1**	**0**